MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE)-MG3DSEBOTA06
PRINCIPLES OF MICROBIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A

Short answer type questions. Answer any 10 questions. Each question carries 2 marks ([2x10=20]

1. Explain why algae are considered primary producers in aquatic ecosystems. [K] / [CO1]

   Photosynthetic energy production, foundation of food webs, global O_2 and CO_2 cycling, Habitat and Water Quality Roles. (Any relevant two points)

2. Name the two photosystems associated with light reaction of photosynthesis. [K] / [CO2]

   Photosystem 1 and Photosystem II

3. Write one advantage of using biofertilizers over chemical fertilizers. [U] / [CO4]

   Biofertilizers enhance soil health by increasing microbial diversity and improving soil structure, thereby promoting long-term soil fertility and sustainability without causing pollution or soil degradation.

4. Why is bacterial morphology important in microbiology? [K] / [CO1]

   Morphology plays a role in bacterial pathogenicity and niche adaptation. Recognizing bacterial shape also aids in the identification and classification of bacterial species.

5. Define fermentation. [K] / [CO4]

Any metabolic process that releases energy from a sugar or other organic molecules, does not require oxygen or an electron transport system, and uses an organic molecule as the final electron acceptor

6. What is the composition of Bacterial cell membrane? [K] / [CO1]

   The membrane is composed of a fluid phospholipid bilayer (40%). Proteins constitute about 60% of the bacterial membrane. Many bacteria contain hopanoids, sterol-like molecules that stabilize the membrane structure.

7. What is transformation in bacteria? [U] / [CO2]

Transformation in bacteria is the process by which bacteria take up free, foreign DNA from their surroundings and incorporate it into their own genetic material. This results in a change in the bacterial genetic constitution, allowing the expression of new traits such as antibiotic resistance or the ability to utilize different nutrients.

8. Why is Gram's staining important in microbiology? [U] / [CO3]

Gram staining allows for the rapid classification of bacteria into two major groups, Gram-positive and Gram-negative, based on their cell wall composition. This differentiation facilitates the identification and diagnosis of bacterial infections, thereby guiding the selection of appropriate antibiotic treatments.

9. State one limitation of SCP uses as food. [U] / [CO4]

The high nucleic acid content in microbial biomass can cause health issues; SCP may provoke allergic reactions in some individuals with sensitive digestive systems. (Any one limitation)

10. Name an example of a phage that follows a lysogenic cycle. [U] / [CO2]

Lambda phage (phage λ), Phage Mu (Any one)

11. What are anaerobic bacteria? [U] / [CO2]

Anaerobic bacteria are microorganisms that can survive and grow in environments with little or no oxygen. They do not require oxygen for their metabolism and produce energy through fermentation or anaerobic respiration.

12. What is a cloning vector? [U] / [CO2]

A cloning vector is a small piece of DNA, such as a plasmid or virus, used in molecular biology to carry a foreign DNA fragment into a host cell where it can be replicated and/or expressed. The vector contains essential features, such as an origin of replication for autonomous replication, a multiple cloning site for inserting foreign DNA, and a selectable marker gene (often antibiotic resistance) to identify cells carrying the recombinant DNA**.**

Part B : Short Essay type questions, <u>answer any 8 questions</u>, each question carries 5 marks [5x8 = 40]

13. Describe the major applications of amylases in the food and textile industries. [K] / [CO4]

Starch hydrolysis in food, baking, brewing, textiles, paper, detergents, clinical and analytical chemistry (Explain any 5 applications in one or two sentences).

14. Define the process of conjugation in bacteria. [K] / [CO2]

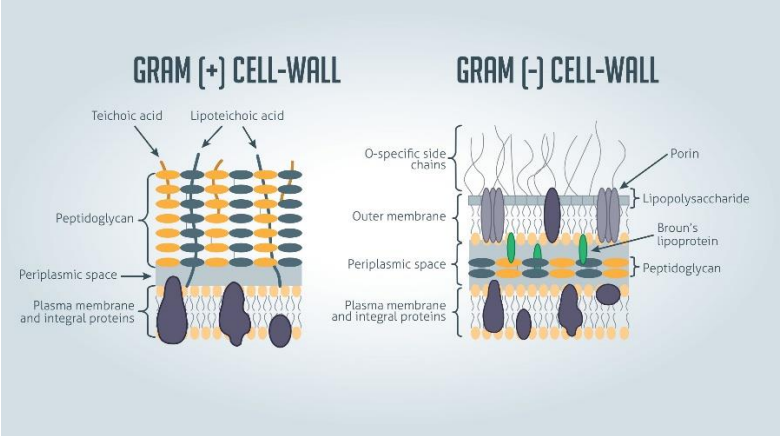
Bacterial conjugation is a process of genetic material transfer between bacterial cells through direct contact. The donor bacterium (F+ cell) forms a sex pilus, a tube-like structure, to connect to the recipient cell (F- cell). The pilus draws the recipient close to the donor, forming a conjugation bridge between their cytoplasm. A plasmid DNA in the donor is nicked by an enzyme at the origin of transfer (oriT). A single strand of plasmid DNA is transferred to the recipient cell through the pilus in the 5' to 3' direction. Both donor and recipient synthesize the complementary DNA strand to restore double-stranded plasmids; the recipient becomes capable of acting as a donor in future conjugations.

15. List the essential elements and growth factors required for bacterial growth, and briefly describe their role. [K] / [CO2]

Carbon, Nitrogen, Phosphorus, **Sulfur, Oxygen, Hydrogen**, **Potassium, Magnesium, Calcium,**.

Vitamins: (e.g. B-vitamin complex, biotin, riboflavin). (Any 5 points and brief explanation)

16. Describe the structure of Gram- negative bacterial cell wall. [K] / [CO1]



A brief explanation or diagram can be considered.

17. Write short notes on endospores. [K] / [CO2]

Endospores are dormant, tough, and highly resistant structures formed by certain Gram-positive bacteria, such as Bacillus and Clostridium, in response to unfavorable environmental conditions. They enable bacteria to survive extreme stresses, including heat, desiccation, chemicals, and radiation.

18. Outline a microbial growth curve [U] / [CO3]

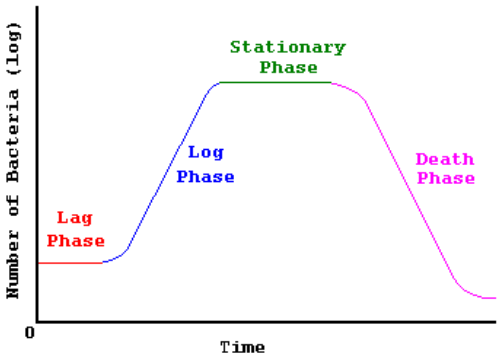


Diagram with brief explanation

19. Write a brief note on culture media used in microbiology. [U] / [CO3]

Culture media in microbiology are nutrient-rich substances used to support the growth, isolation, and identification of microorganisms in vitro. They provide the essential nutrients, moisture, and environmental conditions needed for microbial proliferation.Basal (Basic) Media, Enriched Media, Selective Media, Differential/Indicator Media, Enrichment Media, Transport Media, Storage Media. Physical forms include liquid (broth), solid (agar plates), and semi-solid media, each suited for different laboratory applications, such as cultivation, isolation, or motility testing. (Any 5 types).

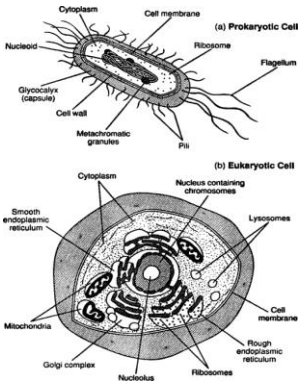
20. Write a brief note on bioremediation [U][CO4]

Bioremediation is an eco-friendly process that utilizes living organisms, such as bacteria, fungi, or plants, to degrade, detoxify, or remove pollutants and contaminants from the environment,

including soil, water, and air. The organisms metabolize the harmful substances, converting them into less toxic or harmless products, such as water, carbon dioxide, and biomass. Bioremediation can be performed in situ (at the site of contamination) or ex situ (removal and treatment elsewhere). This technique is widely used to clean oil spills, pesticides, heavy metals, and other industrial pollutants, promoting environmental sustainability and reducing chemical waste.

21. Draw a labeled diagram of a typical prokaryotic cell and a eukaryotic animal cell. Highlight at least three key differences in structure. [K][CO1]

Prokaryotic Cell	Eukaryotic Animal Cell
Single circular chromosome, no nucleus	Multiple linear chromosomes within a nucleus
No membrane-bound organelles	Membrane-bound organelles present, e.g., nucleus, mitochondria
Smaller 70S ribosomes	Larger 80S ribosomes



22. Differentiate the lytic cycle from the lysogenic cycle of viral replication. [U] [CO2]

Lytic Cycle	Lysogenic Cycle
Viral DNA destroys host DNA, takes over cell functions, and destroys the cell.	Not destroys the cell
The virus replicates and produces progeny phages	The does not produces progeny phages, mitochondria
Symptoms of infection	No symptoms of infection
Virulent viral infection	Temperate viral replication

Part C

Essay-type questions. **Answer any 1 question.** The question carries 10 marks.

(1x10= 10)

23. Write an essay on Glycolysis. [U] [CO2]

Diagram with a brief explanation of each step.

24. Write an essay on the applications of microbiology. [An][CO4]

Any 10 applications with a brief explanation from different areas like Microbial production of enzymes, organic acids, and vitamins; Probiotics & SCPs , Role of microorganisms in agriculture (biofertilizers, biopesticides, biocompost). Environmental microbiology: Waste water treatment and Bioremediation. Fermentation in food production (beer, yogurt, appam, idli).